

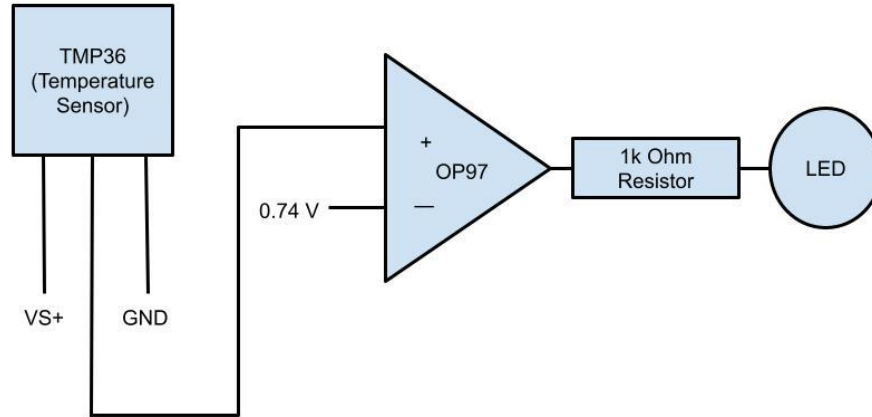
Omega MS1

By Jenna, Carter, and Ryan

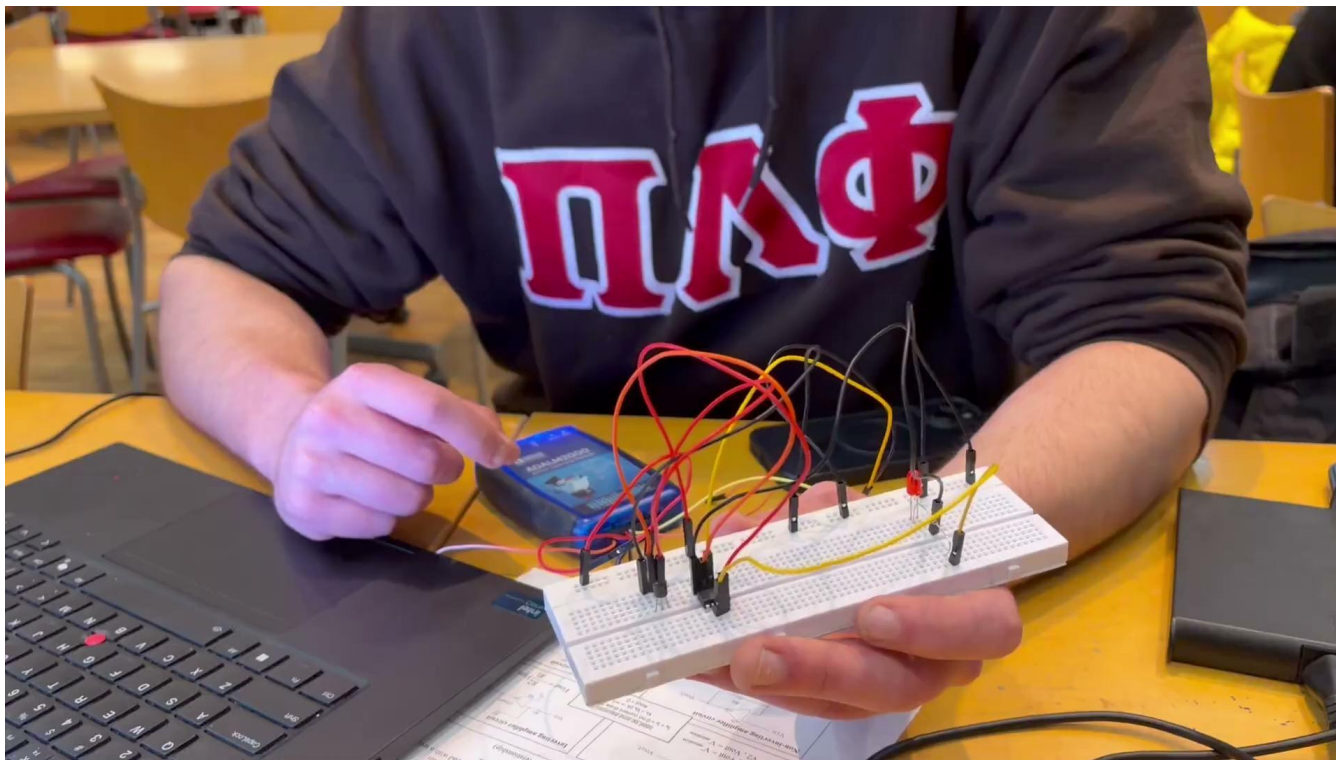
Purpose

The circuit functions as a way to power an output when the detected temperature is reached through the TMP36. Our circuit shows this by powering a LED when the TMP36 outputs a voltage that surpasses the desired temperature voltage through a comparator. The comparator will output a negative voltage when the measured temperature is less than the desired temperature and positive when measured is greater. This positive voltage causes the LED to turn on.

Block Diagram

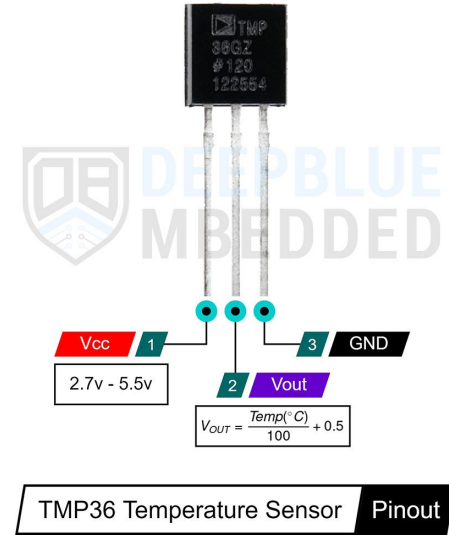


Demonstration



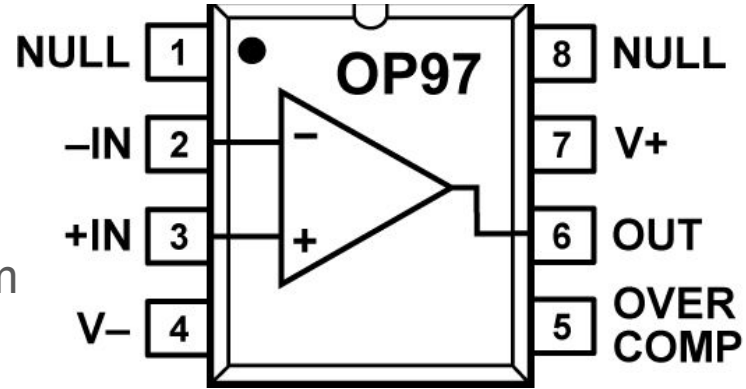
Circuit Operation TMP36

- 3V power supplied to leg 1 of the TMP36
- Vout of the TMP36 (leg 2) wired to the positive supply terminal of the OP97 (pin 3, pictured on next slide)
- GND is supplied to leg 3 of the TMP36
- The TMP36 outputs +10mV per every degree Celsius and has an offset of 0.5V. This means that the equation for Vout is $V_{out} = (0.01\text{Volts}) * (\text{Temperature in Celsius}) + 0.5\text{Volts}$. This means at 24 degrees, $V_{out} = 0.74\text{V}$
- The LED is powered when the TMP36 outputs a voltage greater than 0.74 V`

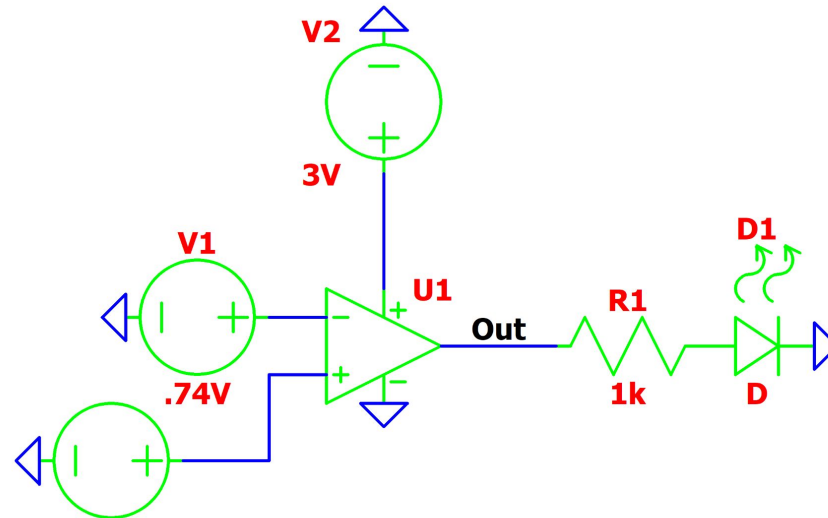
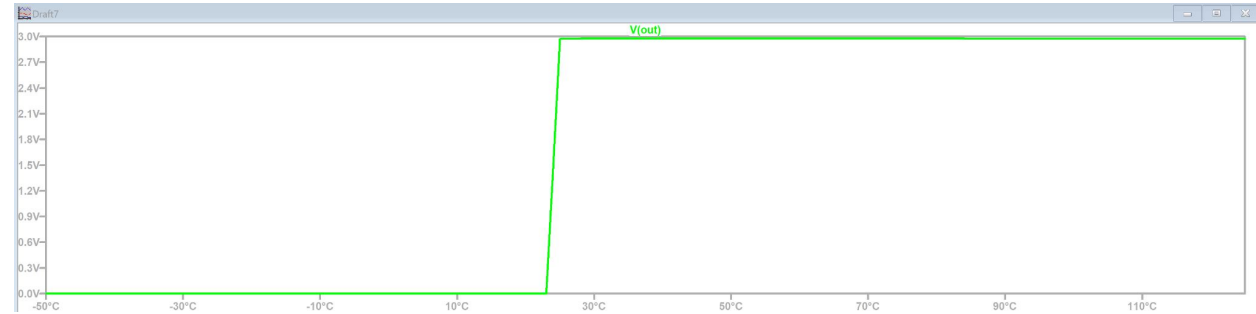


Circuit Operation Op-Amp

- 3V power supplied to pin 7 of the OP97
- The negative supply terminal (pin 2) of the OP97 is wired to W1, set at a threshold of 0.74 V
- GND is supplied to pin 4 of the OP97
- Vout (pin 6) of the OP97 is wired to a 1k Ohm resistor and LED in series
- The LED is lit up when the tmp36 outputs a voltage greater than 0.74 V



Simulation



Temperature_Sensor

$$V=0.01*temp+0.5$$

.step temp -50 125 1

.op

Problems Encountered and Solutions

- One issue we encountered was trouble getting the LED to light up
- After some troubleshooting, we found out that there were some wires in the ground channel of our breadboard that were incorrectly wired, causing the circuit to not be properly grounded
- We needed to use a jumper cable to bridge the gap in the ground channel of the breadboard
- We found this issue through testing different parts of the circuit with a handheld multimeter to see which parts were receiving the intended voltage
- We then rewired the ground channel, which resulted in our circuit working as intended

Design Choices

- TMP36:
 - Voltage that scales with temp
 - Has an offset so the voltage is still positive when the temperature is negative
 - Good price, reliable
 - Proper range of temperatures what we needed for lab (-40°C to $+125^{\circ}\text{C}$)
 - Tolerance for celsius is accurate
- OP97:
 - Simple and safer circuit design
 - No negative supply required
 - We already had it in our parts kit

Plans For Next Milestone(s)

- For Milestone 2 we plan to add a First-Order Passive Filter and Non Inverting Amplifier
 - The First-Order Passive Filter will remove some of the noise created when the temperature is close to the threshold
 - The Non Inverting Amplifier will scale the voltage to a voltage needed for the DC motor to run.
- For Milestone 3 we plan to add a Oscillator Circuit and a First-Order Active Filter
 - An Oscillator Circuit(Astable Multivibrator) will create a continuous square wave that will allow us to control the DC motor's speed proportionally without wasting massive amounts of energy as heat.
 - The First-Order Active Filter will cut down on the noise created by the motor driver. This will create a clean and sharp cutoff of power to the motor